

Advanced Natural Language Processing

Power of LLMs

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Input text

I am hungry and for lunch I want a...

Generative model

Predictive/Classification
model

**Outputs the most
probable next word
given the input
thus far**

sandwich, probability .9
soup, probability .75
car, probability .02

Outputs text

class 1, probability .9
class 3, probability .5
class 6, probability .1

Outputs prediction

**Outputs the most
probable class
the phrase is
aligned with**

The input data represents the content the model will train on. This typically consists of terabytes to petabytes of data.

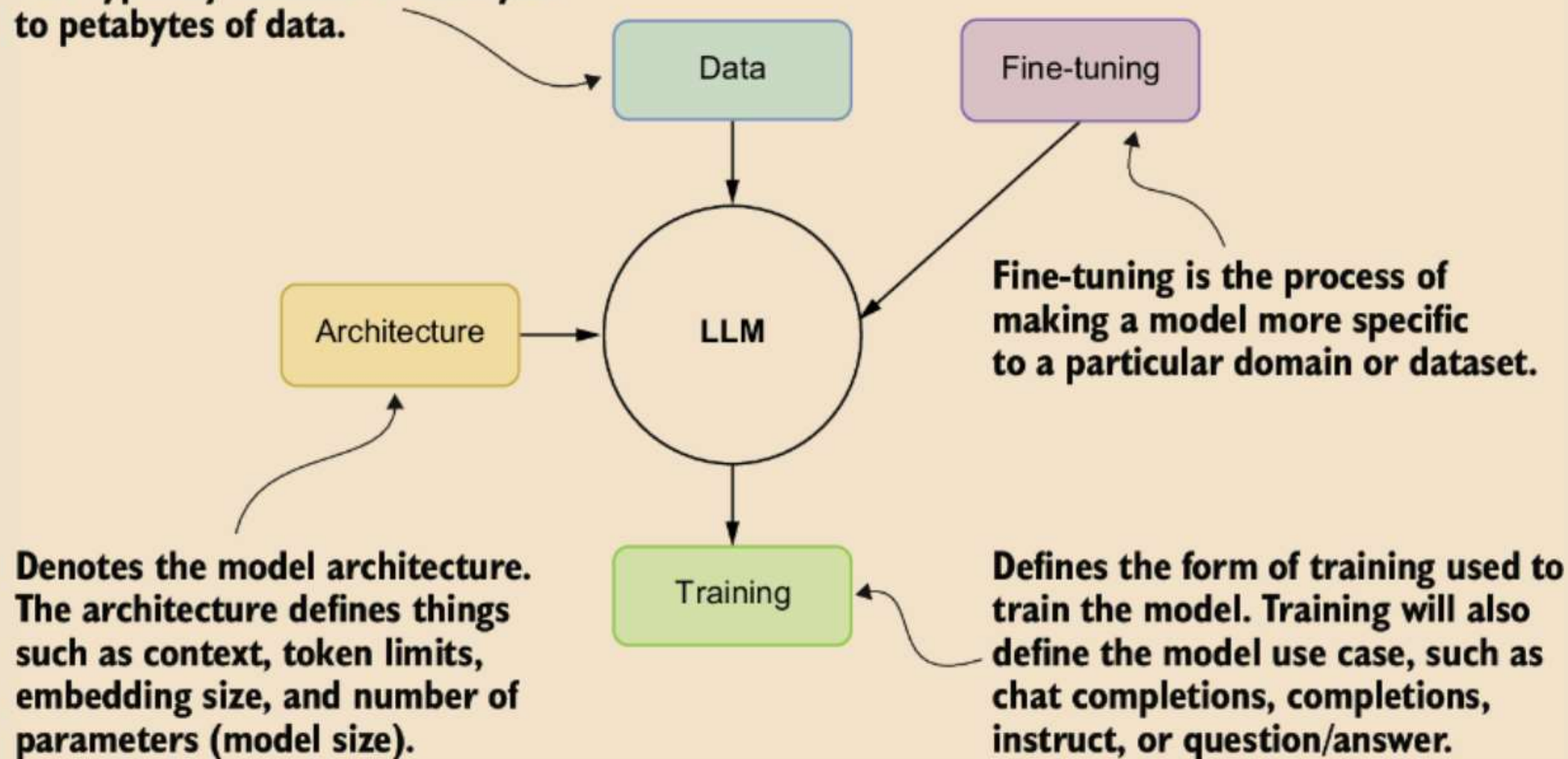


Figure 2.2 The main elements that describe an LLM

Code:

AI_Agent_Book_Ch2_OpenAI.ipynb

- Get a feel of zero shot, one shot and openai API

LM Studio: Running LLMs locally

- Here we will use jupyter notebook (Notebook: lm_studio)

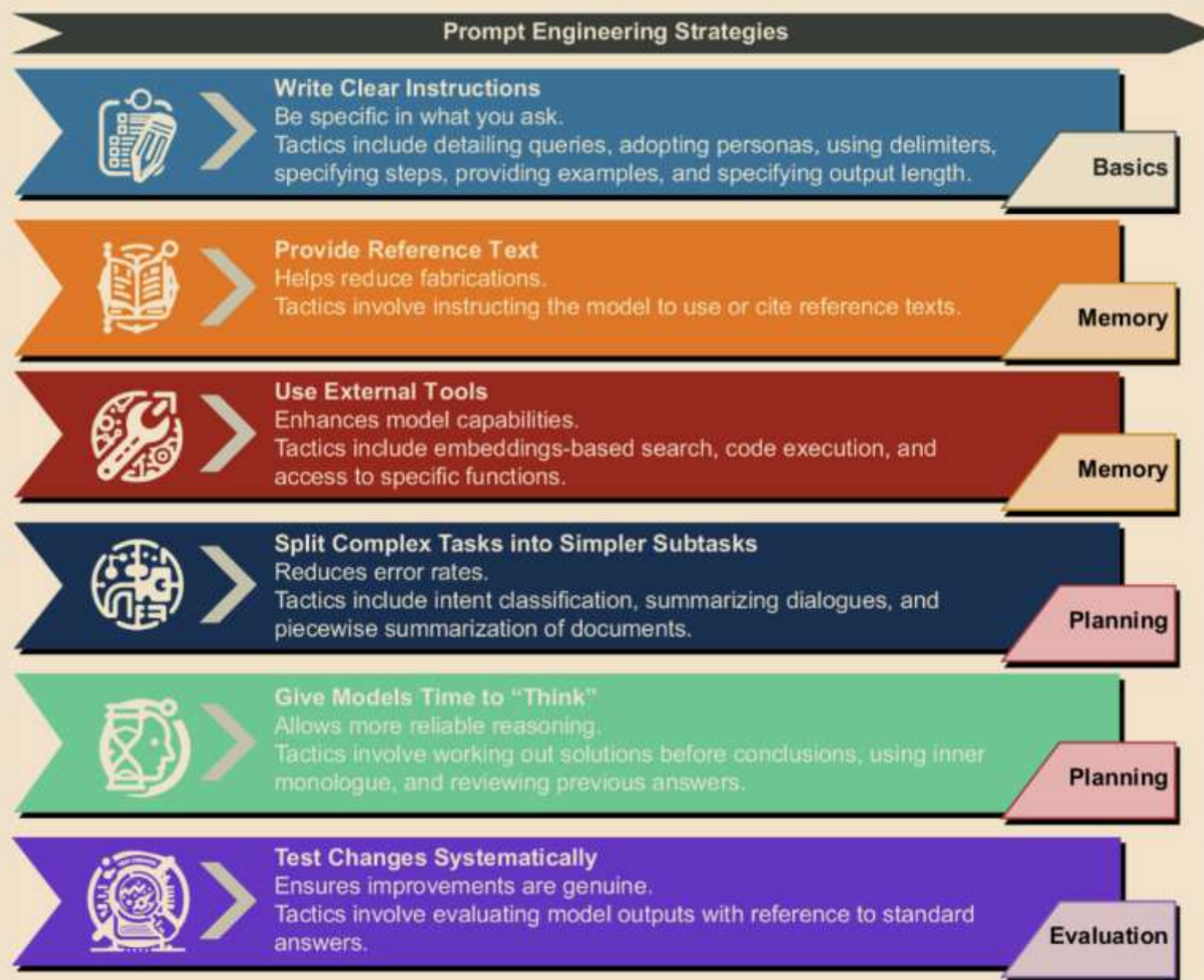


Figure 2.8 OpenAI prompt engineering strategies reviewed in this book, by chapter location

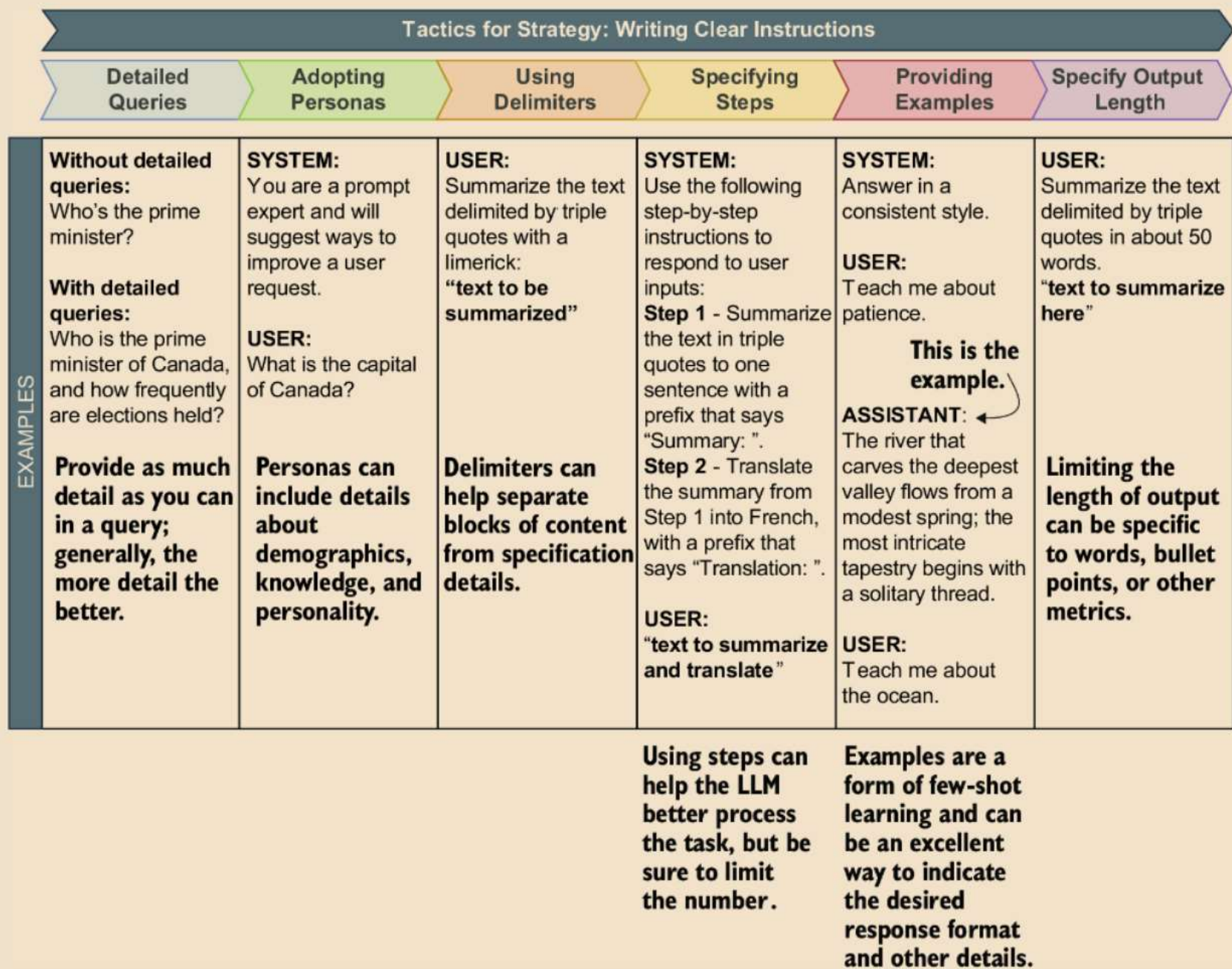


Figure 2.9 The tactics for the Write Clear Instructions strategy

Prompt Strategies

- (Notebook: lm_studio)

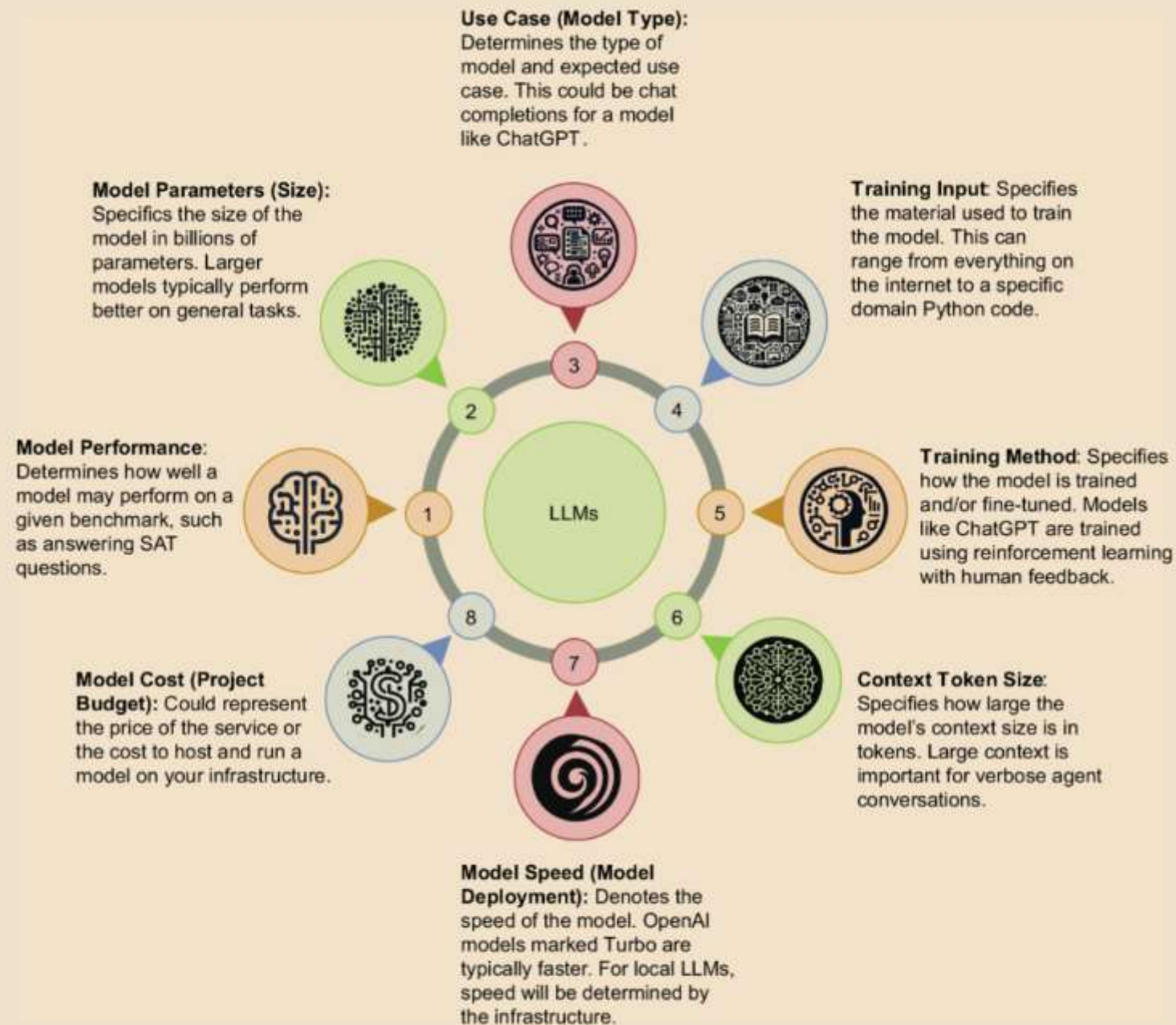


Figure 2.11 The important criteria to consider when consuming an LLM